EAS-5450 Final Exam Review

These slides will be available on your Canvas site

Tuesday, May 6, 12:00-2:00pm
Section 401 (Cassel): BENN 419
Sections 402 (Halak): SKIR AUD
Sections 403 (Babin): DRLB A8

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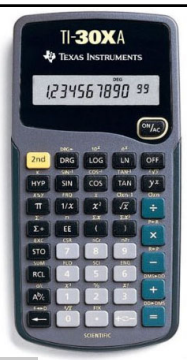
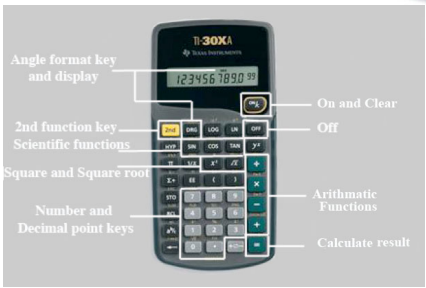
Taking Final Exam – CHARGE YOUR LAPTOP!!!



- Final Exam via Canvas with Respondus
  CONFIGURE & CHARGE your laptop (and ensure it can last at least 2 hours).
  There will be **NO PRINT COPIES AVAILABLE**

- Closed book/readings; you may only use your prepared notes pages.
  Two, 2-sided 8½x11 sheets (4 pages):
  hand-written or copy-and-pasted.
  Must be printed and brought to exam.

- We will provide calculators for use during the exam (model TI-30XA).
  Personal calculators may not be used.
  You may "test drive" a calculator in 304 Towne (Ashley's office) in advance to become familiar with it (recommended).



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Objectives for (short) review session

We will

- Address clarifications of content topics
- Focus on things that will be on the exam, and note those that are not
- Offer occasional case references
- **Answer additional questions as time allows**

We won't

- Rehash the course
- Review cases, case facts, key learnings from cases
- Answer "Questions of the Day" (that's the point of studying 😉)
- Summarize guest speakers' points

The last class summary and Babin's Questions of the Day (next slide) are a great start but are NOT comprehensive.

If it's not on these or your last class summary slides, it may still be on exam.

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EAS5450-2510 Questions of the Day

Additional course topics & concepts, guests

01 Intro
- Who is Babin?
- Why does this course (and the undergraduate minor and graduate certificate) exist in SEAS?
- What happens if I stay?

02 New Venture, HP
- Why is "*Look at the Fish*" the mantra for Penn Engineering Entrepreneurship?
- What made HP a great company?
  (how did HP reflect insights from Stevenson, Drucker, Dyer/Gregersen/Christensen)

03 Winning Strategy, Vermeer
- What are four (or five) sources of cash and how do they differ?
- How can a startup compete effectively in a new market with many large, well-established companies?

04 IP, Palm A
- What are the types of intellectual property and which would you use to protect the IP of your venture? How?

05 Innovation, Curves, and Sales Learning Curve
- How do we turn innovation into products, and products into ventures?
- Why does our venture exist and how does/should it evolve over time?

06 Palm Computing, Crossing the Chasm
- How can a startup succeed where well-established, well-funded companies have failed?
  (HINT: effective marketing!)
- How can you transition from early market visionaries to mainstream pragmatists?

07 Palm Pilot, Customer Discovery
- How can we use cost-effective market research to avoid failure?
  (and to revisit the Palm Pilot)
- *Principles of Concept Test and Conjoint Analysis*

08 Lean Startup, Dropbox
- How does Lean Startup and market research help us (and Dropbox) achieve product-market fit more effectively?

09 LinkedIn
- How does positioning help setup our marketing strategy?
  (foreshadowing of the *Business Model Canvas*)
- *Principles of pricing and impact on decision-making*

10 Dell Virtual Integration, Business Model Canvas, Understanding Financial Modeling & Statements
- What is the impact of a new business model on company, competitive positioning, industry, and financial models?
- *Principles of income statement, balance sheet, cash flow*

11 Dell Product Development, Expected Value
- How can we combine qualitative and quantitative analysis to make better business decisions?

12 Bootstrapping & Raising Capital
- How do VC's evaluate (and not evaluate) an investment opportunity?
- How do high-tech startups get funded?

13 VC Method
- When investing in a startup, how does a VC know how much to pay, how much of the company (shares, %) they get, how much of a return on their investment they will realize, and how they will realize the ROI?
- *Convertible notes, Simple Agreement for Future Equity (SAFE)*

14 Evaluating Term Sheets, Negotiating Skills
- What are the considerations when negotiating terms with venture capitalists and early-stage investors?

15 IPO, Facebook
- Do you want your company to go public?
- How does your answer change if you are: VC investor? Founder, CEO? Founding employee? New employee?

16 Open Innovation, Palm Financing
- How has innovation changed over the past few decades?
- How are investors and entrepreneurs aligned and misaligned when evaluating investment opportunities and liquidity events?
- How are objectives aligned/misaligned among different investors in the same company?

17 GUEST: Ross Mechanic, Maybern

18 Operations, GenapSys
- What is disruptive technology?
- How can disruptive technology and our choice of business strategy and model impact opportunity, cash flows, and funding requirements of hi-tech startup?

19 GUEST: Julia O'Mara, Pickle

20 Nanogene, Founders Dilemma (surprise)
- How should founders determine equity division of ventures?
- What considerations are critical for new hires and equity allocations?
- How can you effectively transition from founders-only to key hires?

21 Sirtris, Open Innovation, FDA
- How are startups in life sciences (and other regulated industries) different from other technology startups?
- How do successful life science startups manage these differences?

22 GUEST: Tobias Dengel, Willow Tree

23 GUEST: Miranda Wang, Novoloop

24 MIPS, Spoiled Startup, Level 5 Leadership
- How can a hi-tech startup compete with an established market-leading company?
- What contributes to success and failure of a hi-tech startup?

25 Sun, First Who…Then What
- What contributes to success and failure of a hi-tech startup?
- [EAS545 review – MIPS & Sun]

26 Motivation, Scaling a Startup, N12
- What are considerations for scaling a successful high-tech startup?
- How can you effectively motivate employees?

27 Theranos, Principled Entrepreneurship
- How can well-intentioned entrepreneurs lead a company so far astray of ethical, moral boundaries?

28 Leaders & Managers, Creators & Stewards
- How do I prepare for the final?
- What did I learn (should I have learned) this semester?

Final Exam Do's and Don't's

You should:

- Focus on key concepts (especially those referenced multiple times in classes)
- Understand and be able to apply key concepts
- Know key cases and learnings, especially those referred to repeatedly
- Cite case companies & players, as well as guests, as examples of your responses (with some connection to the actual case/guest)
- Realize that successful entrepreneurs and hi-tech ventures are a balancing act of managing the uncertainty of developing product, customer, market, operations, funding, and team, NOT following a recipe.

We will not:

- Require you to know details (numbers, dates, etc.) from all cases
- Ask questions about the epilogue of cases
  (in fact, the epilog may be irrelevant to the actual learnings of the cases)
- Explain the key takeaway from each case or guest lecture

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Topics addressed today in response to 2(!) questions submitted

- Value proposition (Complete Products, Business Model Canvas)
- Product-Market Fit (Lean Startup, Hypothesis-Driven Entrepreneurship)

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Create products not devices to deliver value

Devices ⇒ Complete Products

What we invent	What customers buy
Xxx-based widget that is capable of geeking out faster than Yyy-based thingamajig	*Value, benefits, experience* $ ⏲ ☺/☹

A product is the totality of what a customer buys.
William Davidow, *Marketing High Technology*

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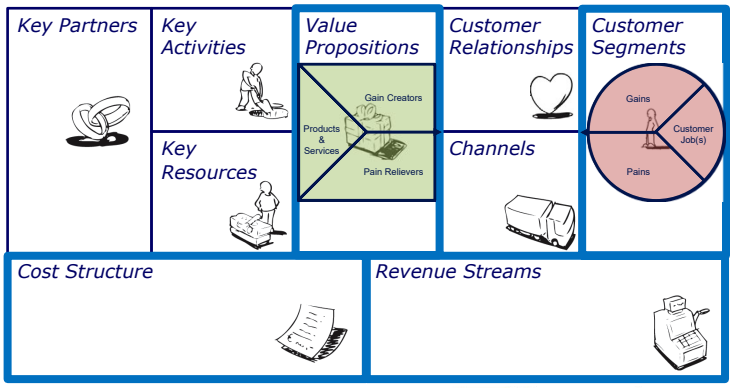
Business models reflect the key choices you make in elements of business

- Business Model Canvas presented 9 elements in framework
  Value Proposition Canvas focuses on the 2 most important: CS, VP
- Product-Market Fit is key to POSITIONING (=segmentation + differentiation)
- Value Proposition solves a customer problem or satisfies a customer need.
- *Innovating for Cash* identified three key business models for operations
  - Integrator
  - Orchestrator
  - Licensor

What? Why? How?



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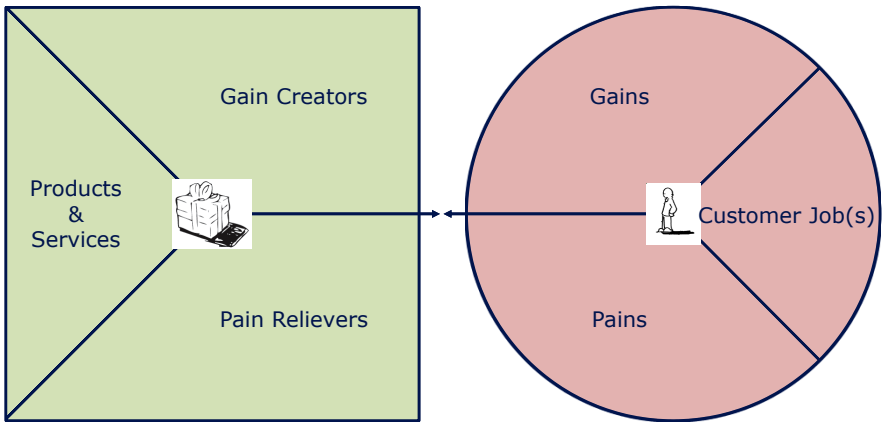
Achieve product-market fit (then scale as fast, large as possible)

The Value Proposition Canvas



Adapted from the Value Proposition Canvas, http://www.businessmodelalchemist.com.

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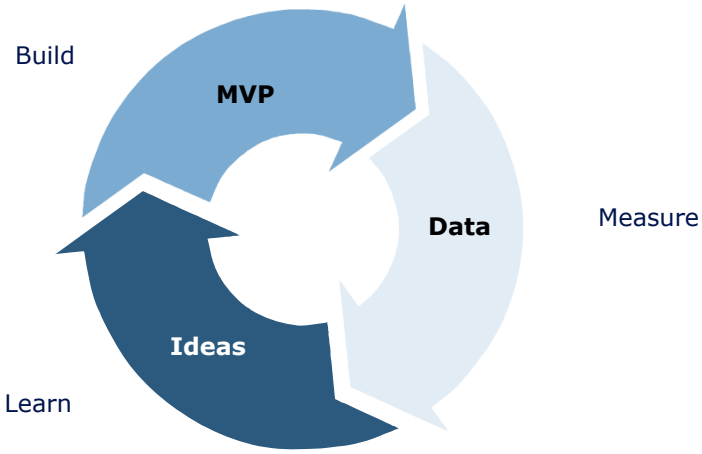
The Lean Startup Model minimizes time in the Build-Measure-Learn feedback loop



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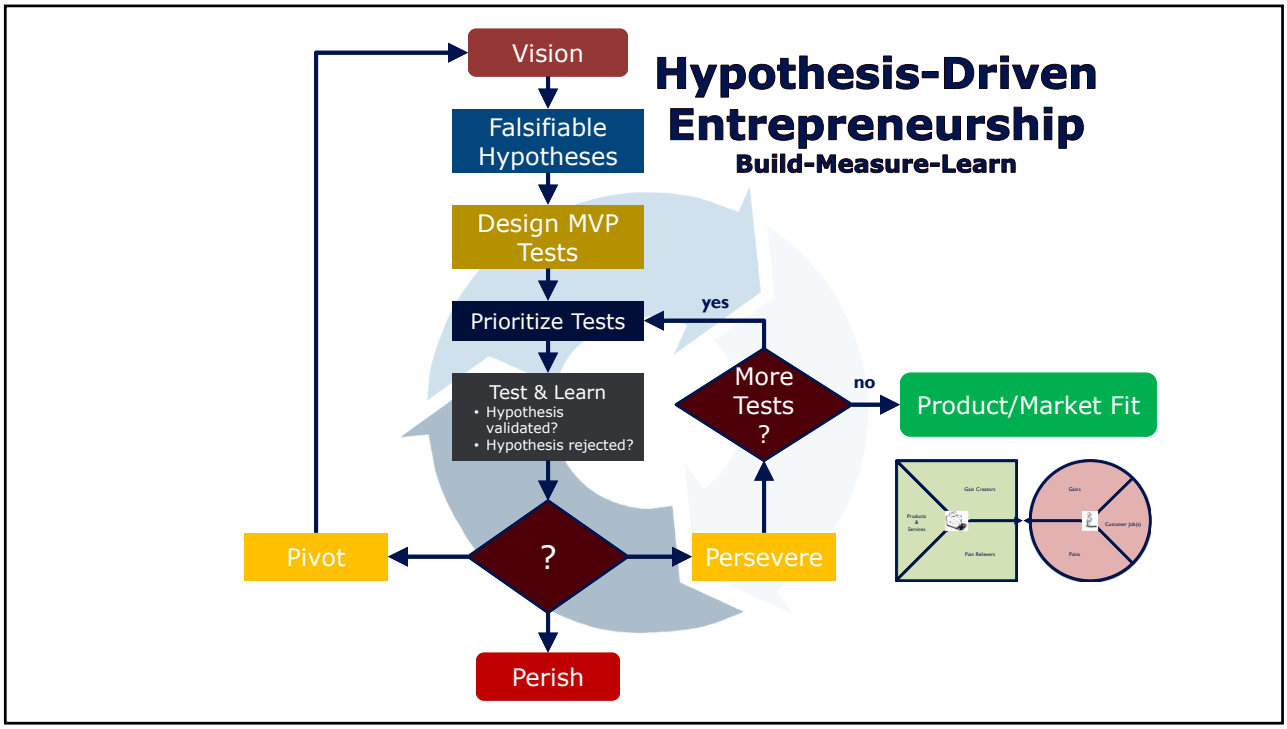

Good Luck on the Final!

For Babin's section: Slack #final-exam

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